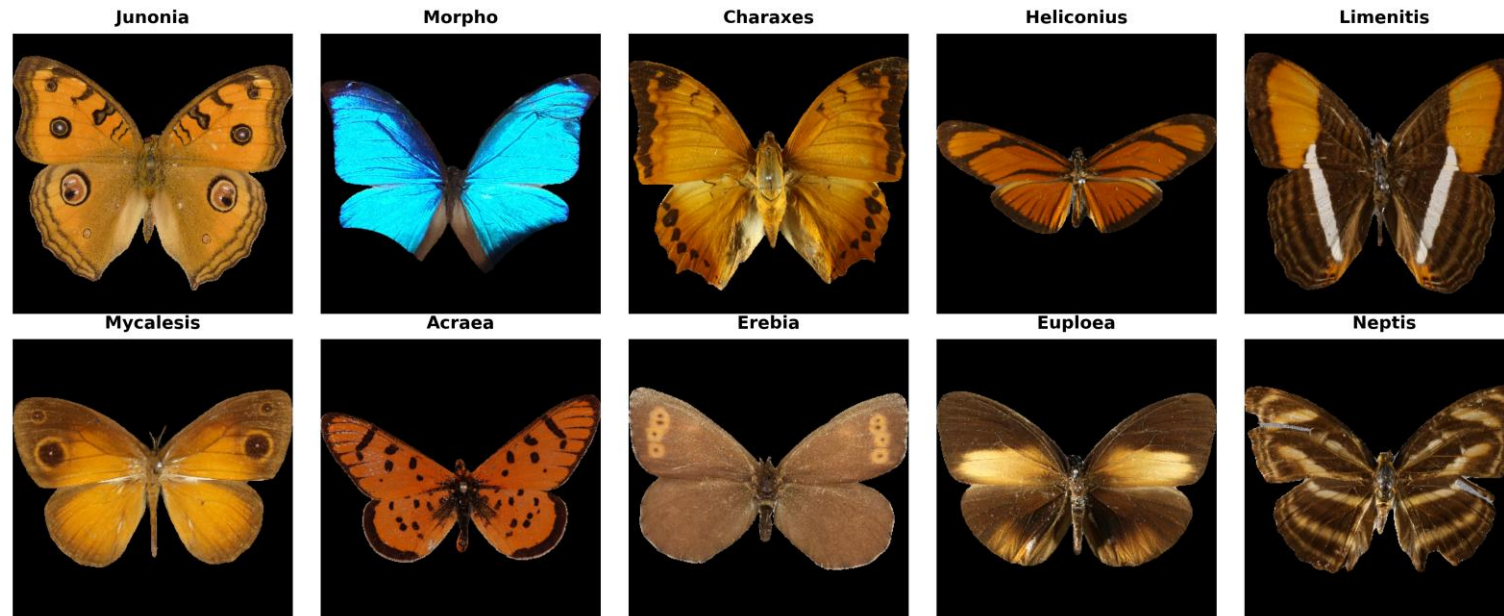


Self-Supervised Vision Embeddings Reveal Phenotype–Taxonomy Structure in Digitized Butterfly Collections

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"Can self-supervised visual embeddings recover species structure—and reveal biologically meaningful departures from taxonomy?"

MOTIVATION & OPERATIONAL BOTTLENECK

Millions of Specimen Images, Limited Expert Review Capacity



Digitization Acceleration

Biological image archives are expanding far faster than manual expert curation can scale.



Repository noise

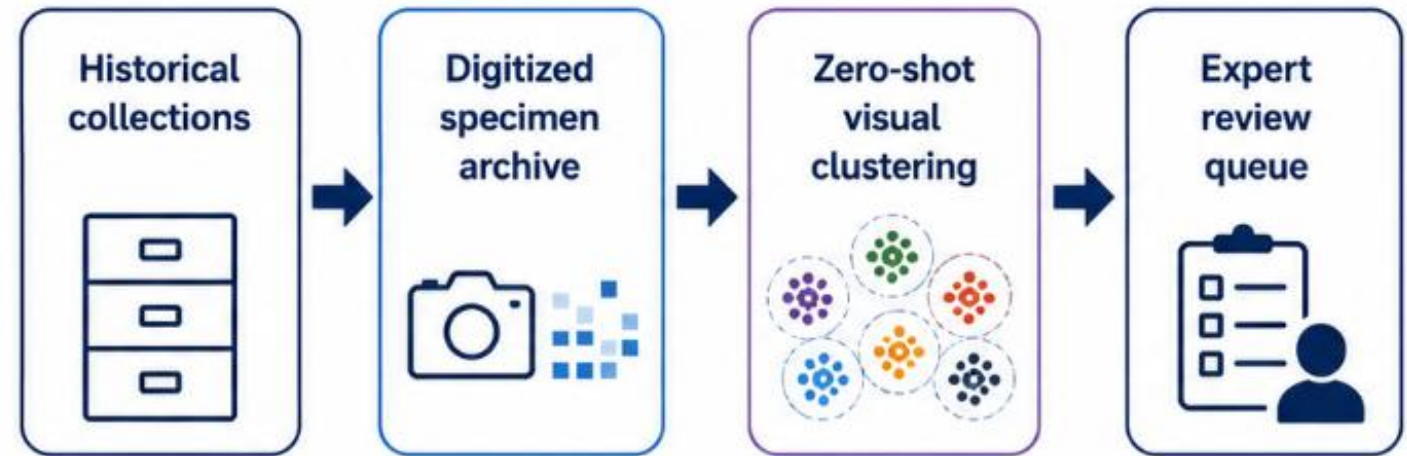
Historical collections contain unresolved boundaries, labeling errors, and subtle population structures.



Supervised Trap

Supervised classifiers force images into predefined species labels, inheriting training label noise and failing on novel or variable morphs.

Proposed Approach



Why this helps



Label validation



Collection audit



Retrieval



Review prioritization

Visual clustering offers a practical triage tool for large digitized natural history collections

CONCEPTUAL FRAMEWORK

Taxonomy-Free Visual Clustering: Emergence vs. Prediction

Phase 1: Pure Visual Structuring

Zero-Shot Representation:

- No taxonomic labels used during feature extraction.
- No labels used for dimensionality reduction (PCA/t-SNE/UMAP).
- No labels used for partition or density clustering.

Phase 2: Post-Hoc Taxonomic Audit

External Audit Signal:

- Ground-truth species names introduced *only afterward*.
- Quantifies alignment (ARI, V-Measure) vs. geometry (Silhouette).
- Diagnoses biologically informative departures.

We do not train the model to predict taxonomy; we ask whether taxonomy naturally emerges from pure visual structure.

Why Butterflies?

A natural intersection of biological complexity and computer-vision difficulty

Natural history / Biological significance

- Wing patterns encode mimicry, polymorphism, geography, seasonality, and divergence.
- Species labels and visual phenotypes are related but non-equivalent.
- Standardized specimen images retain provenance and taxonomic context.
- Mismatches can guide expert review and collection triage.



DIVERSITY • COMPLEXITY • AMBIGUITY

Computer vision significance

- Fine-grained differences and within-species variation make recognition hard.
- Open-set structure exposes limits of fixed-label classification.
- Long-tailed sampling reflects real biodiversity archives.
- Enables clustering, retrieval, anomaly detection, and representation analysis.

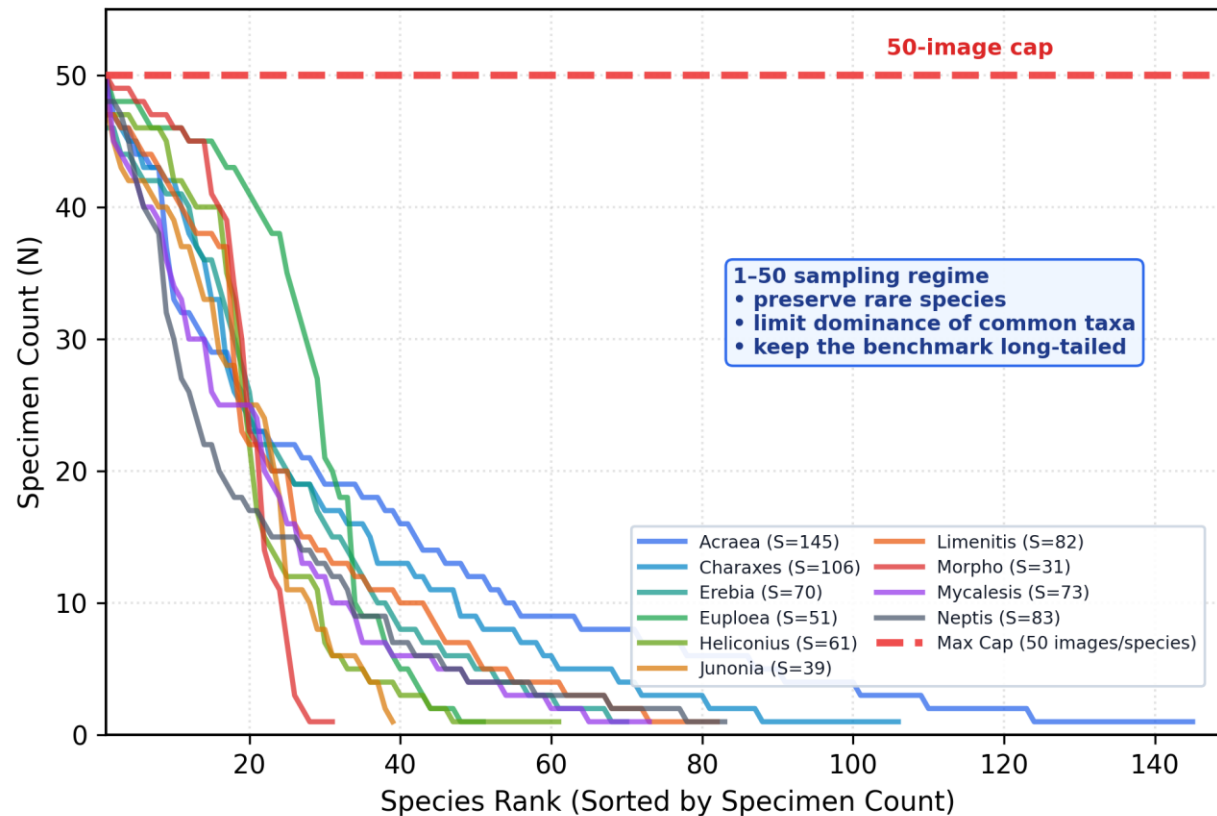
Butterflies are visually rich, biologically interpretable, and computationally challenging – an ideal testbed for taxonomy-free visual representations.

DATASET & BENCHMARK CONSTRUCTION

Species-capped 1–50 Sampling Preserves Long-Tailed Collection Structure

230,000+
SOURCE IMAGES189,541
DORSAL ARCHIVE11,821
BENCHMARK IMAGES741
RECOGNIZED SPECIES

Long-Tailed Species Support

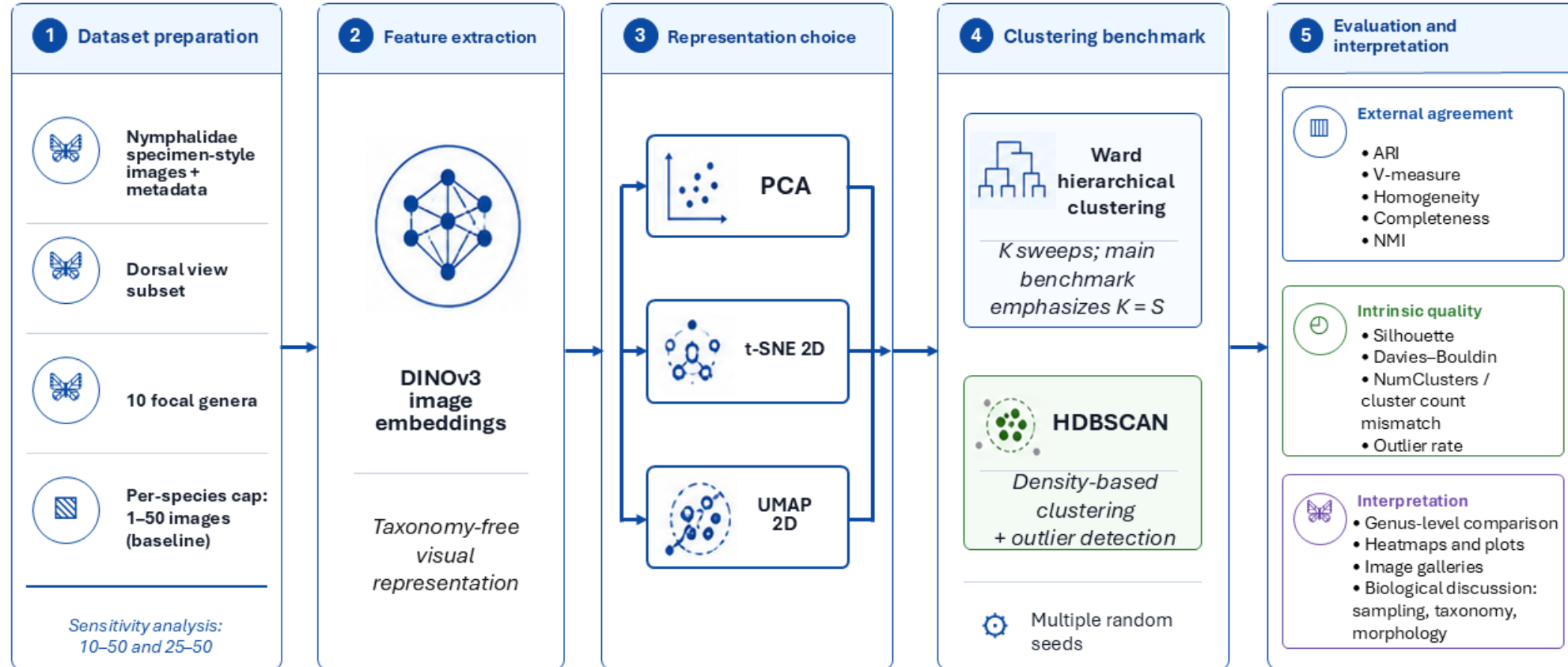


Genus	S	N	N/S
Junonia	39	919	23.56
Morpho	31	943	30.42
Charaxes	106	1,403	13.24
Heliconius	61	1,020	16.72
Limenitis	82	1,246	15.20
Mycalesis	73	1,030	14.11
Acraea	145	1,675	11.55
Erebia	70	1,206	17.23
Euploea	51	1,378	27.02
Neptis	83	1,001	12.06

1–50 sampling keeps rare species while preventing abundant species from dominating the evaluation.

Zero-Shot Benchmarking Pipeline

Images → DINOv3 → PCA / t-SNE/UMAP → Ward / HDBSCAN → Evaluation



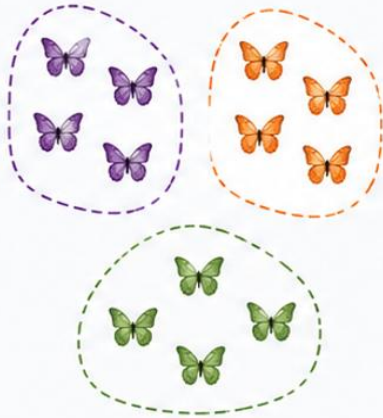
Goal: Assess whether DINOv3 visual embeddings recover species-consistent structure across genera, and diagnose when it does not.

Zero-Shot Benchmarking : Evaluation - Extrinsic & Intrinsic

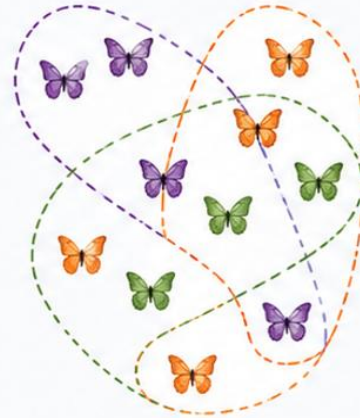


ARI: Taxonomic agreement

High agreement



Low agreement



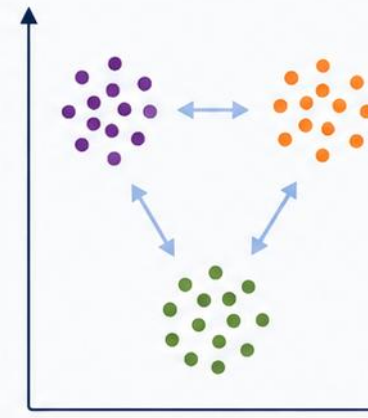
cluster-label agreement

higher is better

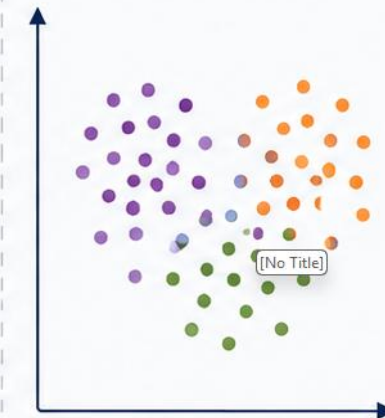


Silhouette: Geometric separation

Good (high silhouette)



Poor (low silhouette)



compact + separated

higher is better

Species labels (ground truth)



Species A



Species B



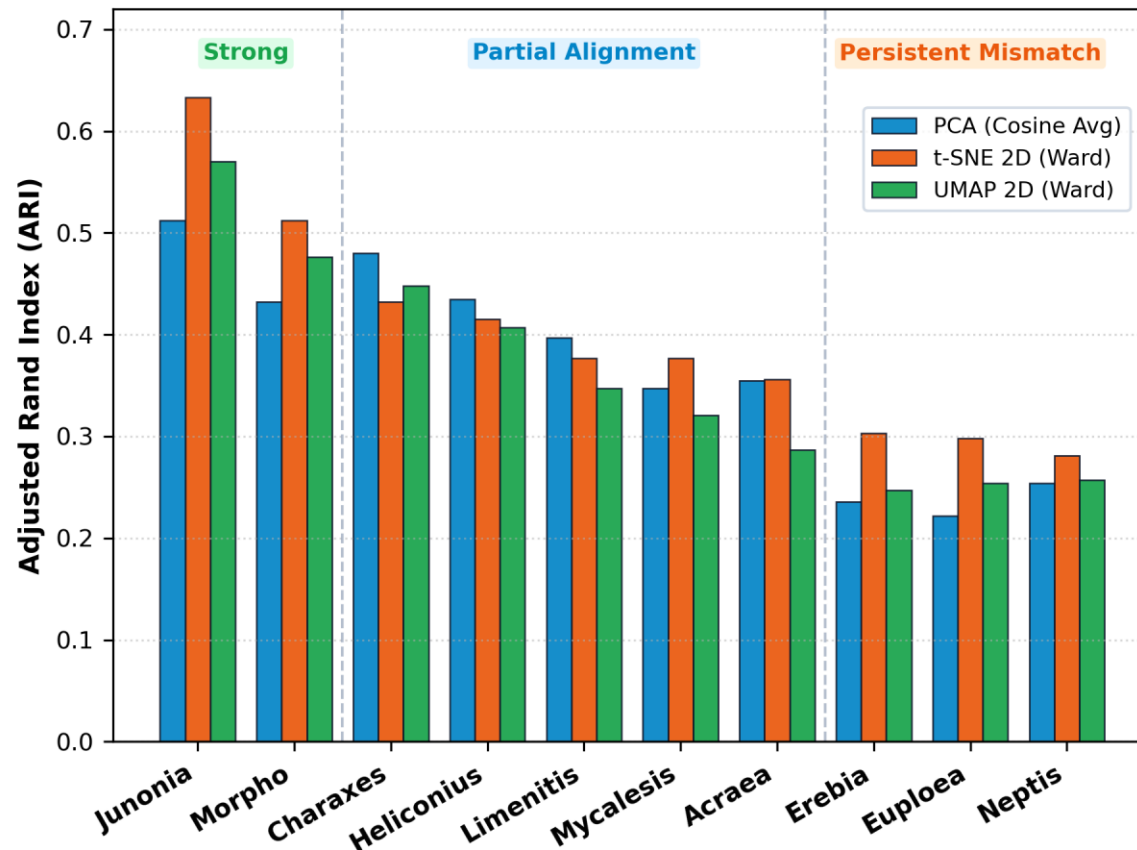
Species C

Predicted clusters



Genus Identity Matters More Than Representation Method

Best ARI by Genus Across Representation Spaces



Strong Alignment

Junonia (0.645) & Morpho (0.512): Discrete, visually striking wing patterns align tightly with species boundaries.

Partial Alignment

Charaxes, Heliconius, Limenitis: Coherent visual clusters split nominal species due to dimorphism or mimicry.

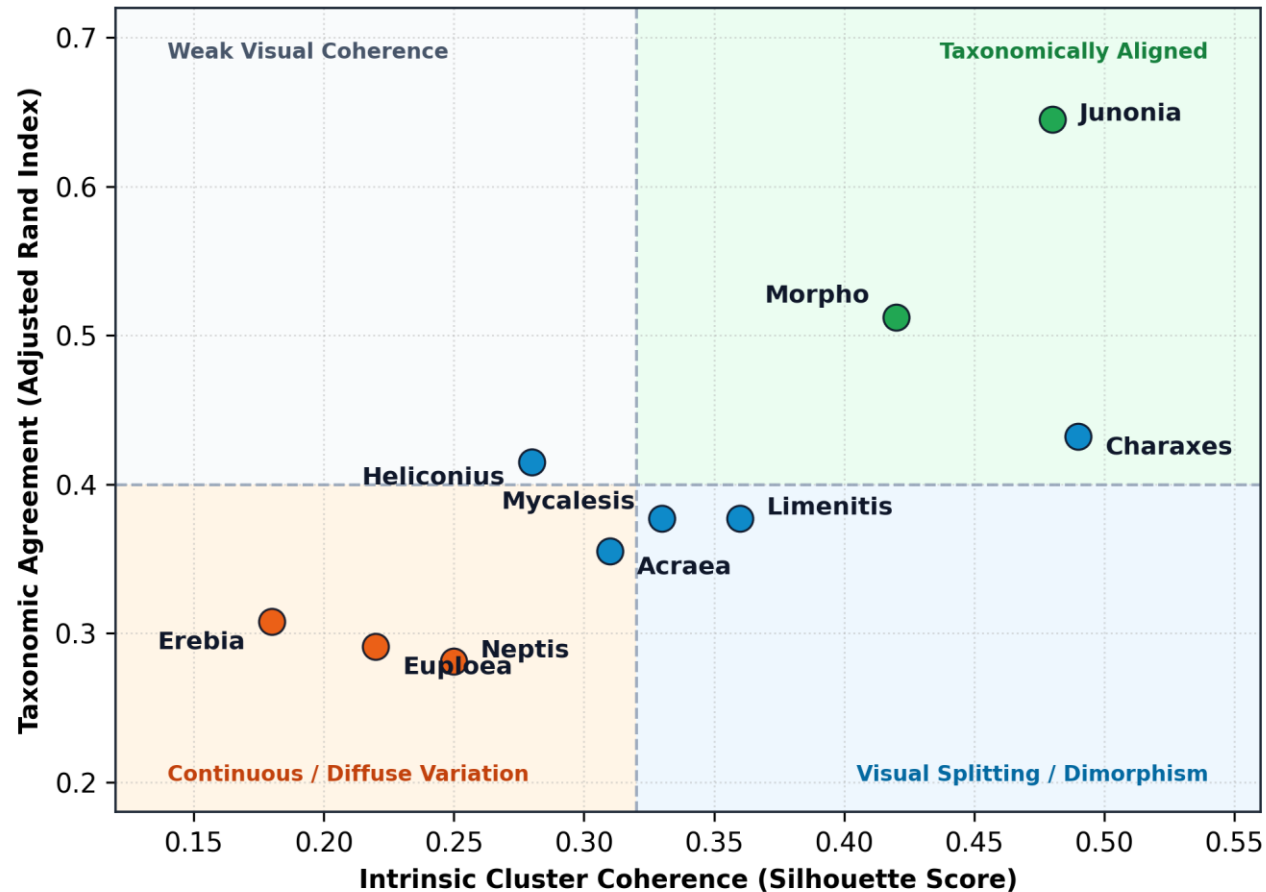
Persistent Mismatch

Erebia (0.308) & Euploea (0.291): Continuous or convergent phenotypes cause persistent visual mixing across taxa.

Visual clustering performance is shaped by evolutionary biology and phenotypic structure, not only by dimensionality reduction.

A Visually Good Clustering Is Not Necessarily Taxonomically Correct

Intrinsic Quality vs. Taxonomic Agreement



Taxonomically Aligned

Junonia & Morpho - High ARI, High Silhouette: Visually discrete species boundaries. Visual geometry matches taxonomy.

Visual Splitting / Dimorphism

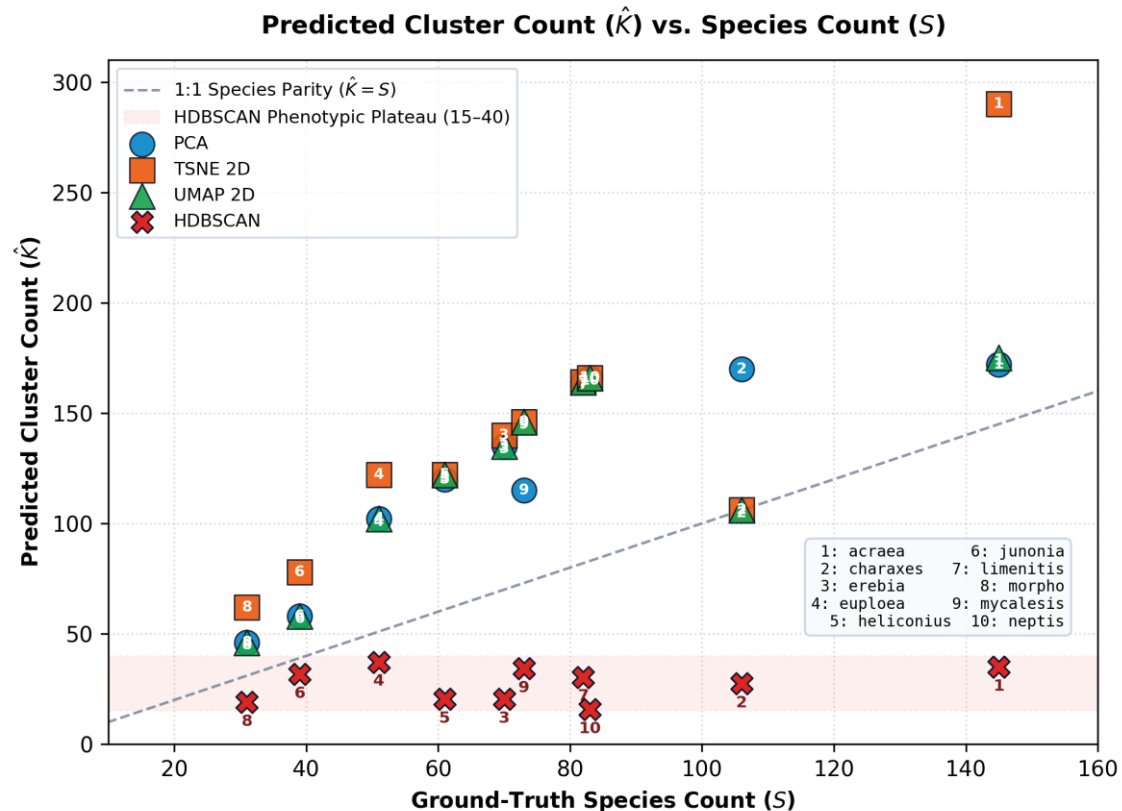
Charaxes, Heliconius, Limnitis - Moderate ARI, High Silhouette: Coherent visual groups that cut across species. Form dense visual clusters, but split nominal taxa due to sexual dimorphism.

Continuous / Diffuse Variation

Erebia, Euploea & Neptis - Low ARI, Low Silhouette: continuous, conserved, or convergent visual morphospace; weak species-level separability.

High cluster coherence does not guarantee taxonomic agreement—unsupervised geometry isolates biological morphospace (e.g., dimorphism, mimicry) rather than nominal labels.

Dense Visual Archetypes Are Often Coarser Than Recognized Species



15 – 40 mode ceiling

HDBSCAN tends to recover a smaller envelope of dense visual modes, roughly 15 to 40 per genus

Phenotype Compression

In uniform or mimetic clades (*Neptis*, *Acraea*), multiple nominal species converge onto single shared dense archetypes.

Outlier Noise Isolation

HDBSCAN autonomously isolates 3.5%–16.7% of specimens as unclustered noise (rare morphs, damaged specimens, atypical lighting) rather than forcing them into artificial groups.

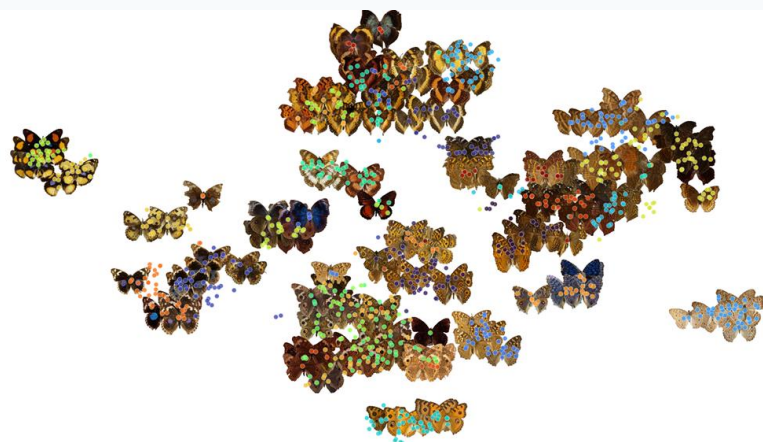
Density clustering recovers a smaller set of high-density visual modes, suggesting that dominant visual structure is often coarser than species-level taxonomy.

QUALITATIVE EMBEDDING ANALYSIS

Visual Embedding Landscapes Across Phenotypic Regimes

Strong Alignment

Junonia



Discrete visual neighborhoods track species labels directly in image space.

Partial Alignment

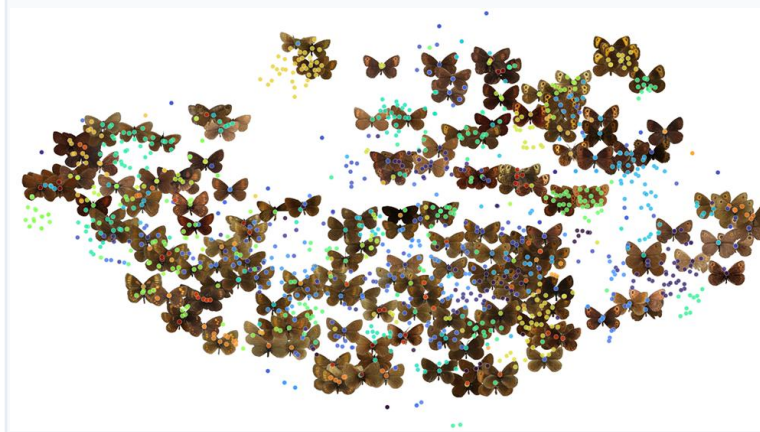
Charaxes



Coherent neighborhoods form, but sexual dimorphism splits nominal species.

Weak / Diffuse

Erebia



Continuous visual variation leads to weak species-level separability.

Biological Drivers: Disagreement is not automatically a model failure—it arises from polymorphism, Müllerian mimicry, sexual dimorphism, seasonal polyphenism, geographic variation, or historical curation noise.

Toward Collaborative Natural-History AI

What this benchmark enables

- A taxonomy-free way to organize large specimen archives by visual similarity.
- A triage workflow for fragmented taxa, mixed clusters, atypical specimens, and label inconsistencies.
- A reproducible benchmark for evaluating foundation-model representations in biological collections.

Future work / collaboration opportunities

- **Metadata + phylogeny grounding:** integrate evolutionary, spatial, temporal, ecological, and provenance context.
- **High-dimensional density discovery:** identify phenotypic modes directly in foundation-model space.
- **Cross-collection validation:** test robustness across museums, taxa, imaging standards, and model families.

Open Reproducibility Artifacts Released: Metadata, dorsal images, run-level result tables, precomputed DINOv3 embeddings, and notebooks.

Looking to connect: biodiversity collections • multimodal foundation models • graph learning • open-set evaluation